

Modified Gravity with Boundary Terms:

A novel geometric modification of General Relativity

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Outline

- 1 Introduction
- 2 General Relativity
- 3 Gamma Squared Action
- 4 Modified Gravity
- 5 Applications
- 6 Summary

Why modify gravity?

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- Dark energy
Cosmological constant problem, coincidence problem, Hubble tension.

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Cosmological constant problem, coincidence problem, Hubble tension.
- Dark matter
Coincidence problem (again), small-scale structure discrepancies.
- Quantum Gravity
Non-renormalizability, singularities, etc...

Modifications of gravity

How do we modify GR?

Modifications of gravity

Lovelock's theorem (1971): *Only second-order, divergence-free, symmetric, tensorial^{*} equations constructed solely from the metric in $4D$ are $G_{\mu\nu} + \Lambda g_{\mu\nu}$.*

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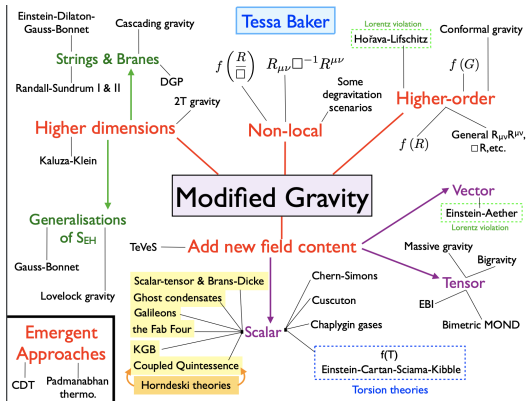


Image credits: Dr Tessa Baker, <http://www.tessabaker.space/>

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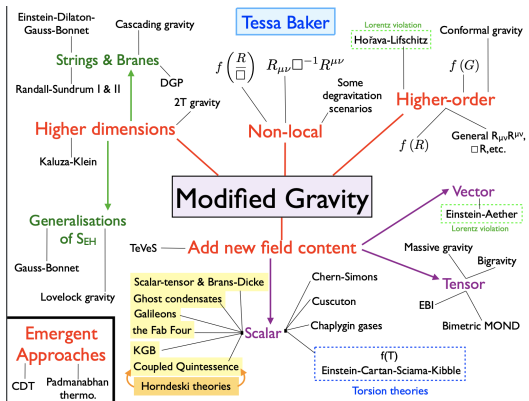


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Gamma Squared Action
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Modified Gravity
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Applications
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Summary
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Einstein-Hilbert action

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- EH action

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$$R_{\mu\nu\gamma}{}^{\lambda} = 2 \overbrace{\Gamma_{\gamma[\mu}^{\rho} \Gamma_{\nu]\rho}^{\lambda}}^{\propto \partial g} + 2 \overbrace{\partial_{[\mu} \Gamma_{\nu]\gamma}^{\lambda}}^{\propto \partial^2 g}$$

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- Include matter $S_M[g, \Phi]$

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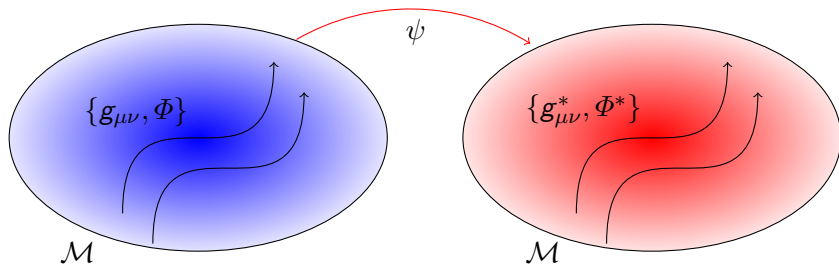
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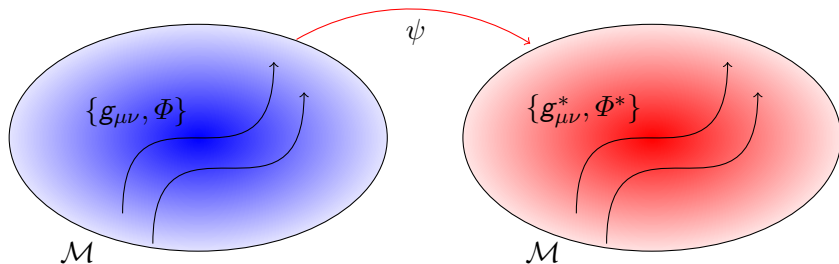
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Diffeomorphism $\psi : \{\mathcal{M}, g_{\mu\nu}, \Phi\} \rightarrow \{\mathcal{M}, g^*_{\mu\nu}, \Phi^*\}$

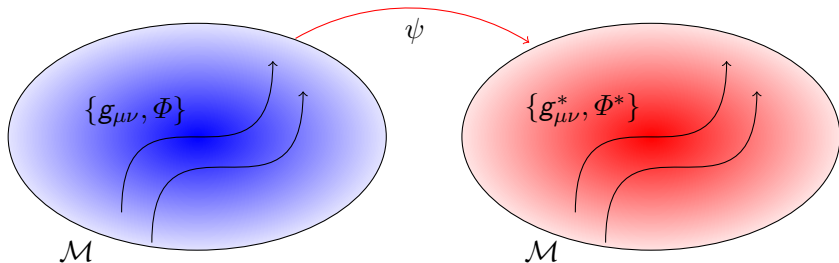
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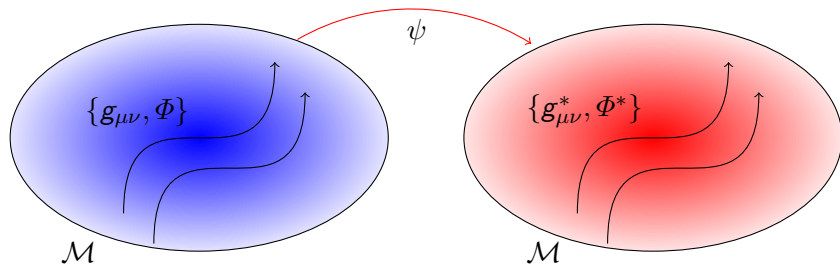
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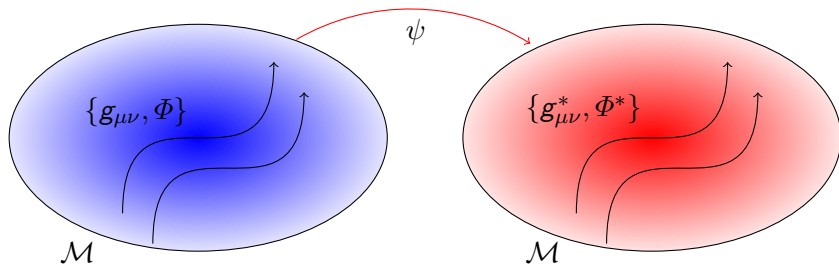
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- Diffeomorphism $\approx$ coordinate transformation

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Gamma Squared Action
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- Lie derivative of metric $\implies$ spacetime symmetries

$$\mathcal{L}_\xi g_{\mu\nu} = 2\nabla_{(\mu} \xi_{\nu)} = 0$$

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- For any scalar action $S = \int \sqrt{-g} L d^4x$

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with ξ held fixed at the boundary.

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- Covariant conservation of energy-momentum $\nabla^\mu T_{\mu\nu} = 0$.

Ricci scalar decomposition

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- Einstein-Hilbert Lagrangian density

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- $\mathbb{B}$ is a boundary term:

$$\mathbb{B} = \frac{1}{\sqrt{-g}} \partial_{\sigma} (\sqrt{-g} B^{\sigma}) \simeq \nabla_{\sigma} B^{\sigma}$$

where $B^{\sigma} = g^{\mu\nu} \Gamma_{\mu\nu}^{\sigma} - g^{\sigma\nu} \Gamma_{\lambda\nu}^{\lambda}$

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- $\mathbb{B}$ does not contribute to EoM!
 $\implies$ “EoM from $\sqrt{-g}R$ ” = “EoM from $\sqrt{-g}\mathbb{G}$ ”
- Consider new action

$$S_{\text{E}} = \frac{1}{2\kappa} \int \sqrt{-g} \mathbb{G} d^4x$$

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Gamma squared or Einstein action:

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- Action *is* diffeomorphism invariant!

Diffeomorphism invariance

Diffeomorphism invariance

- Lie derivative of $\mathbb{G}$ and $\mathbb{B}$:

$$\mathcal{L}_\xi \mathbb{G} = \xi^\mu \partial_\mu \mathbb{G} + M^{\mu\nu}{}_\lambda \partial_\mu \partial_\nu \xi^\lambda$$

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- First order in metric derivative
- Symmetric over $\mu\nu$
- More fundamental than initially looks!

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- Infinitesimal diffeomorphism of S_E

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 &= \frac{1}{2\kappa} \int \left[\underbrace{\cancel{\partial_\mu(\sqrt{-g}\xi^\mu \mathbb{G})}}_{\text{boundary}} + \sqrt{-g} M^{\alpha\beta}{}_\gamma \partial_\alpha \partial_\beta \xi^\gamma \right] d^4x
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- Integrate by parts twice, ξ and derivatives held fixed on boundary

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Introduction
○○

General Relativity
○○○○○

Gamma Squared Action
○○○○○

Modified Gravity
●○○○○○○○

Applications
○○○

Summary
○

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- Neither is $T_{\mu\nu}$ (so S_{M} is not assumed to be scalar).

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- Will impose extra set of conditions along with field equations
- Will see they're required for mathematical consistency (from field equations themselves)

Introduction
○○

General Relativity
○○○○○

Gamma Squared Action
○○○○○

Modified Gravity
○○●○○○○

Applications
○○○

Summary
○

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- Does *not* vanish in general, unlike the Einstein action case

Introduction
○○

General Relativity
○○○○○

Gamma Squared Action
○○○○○

Modified Gravity
○○○○●○○○

Applications
○○○

Summary
○

Conservation equations

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- Can also satisfy equation for *specific coordinates and metrics!*

Introduction
○○

General Relativity
○○○○○

Gamma Squared Action
○○○○○

Modified Gravity
○○○○●○○

Applications
○○○

Summary
○

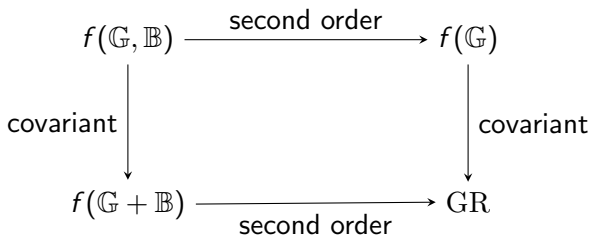
$f(\mathbb{G})$ gravity

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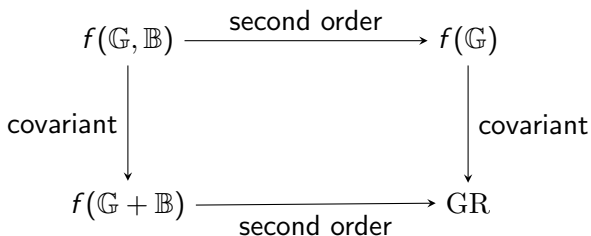
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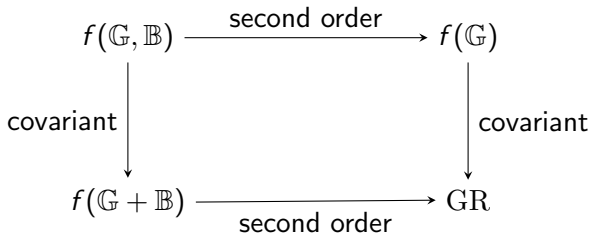
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$$f'(\mathbb{G}) \left[\underbrace{G_{\rho\sigma}}_{\propto \partial^2 g} + \frac{1}{2} g_{\rho\sigma} \mathbb{G} \right] + \frac{1}{2} f''(\mathbb{G}) \underbrace{E_{\rho\sigma}{}^\gamma}_{\propto \partial g} \underbrace{\partial_\gamma \mathbb{G}}_{\propto \partial^2 g} - \frac{1}{2} g_{\rho\sigma} f(\mathbb{G}) = \kappa T_{\rho\sigma}$$

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$f(\mathbb{G})$ field equations

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- (Continuing work): show explicitly that these two equations are mathematically equivalent

Introduction
○○

General Relativity
○○○○○

Gamma Squared Action
○○○○○

Modified Gravity
○○○○○○○○●

Applications
○○○

Summary
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- Matter is assumed to couple to gravity in some unusual way (i.e. to $\mathbb{G}$ and $\mathbb{B}$ such that $T_{\mu\nu}$ isn't a tensor)

Introduction
○○

General Relativity
○○○○○

Gamma Squared Action
○○○○○

Modified Gravity
○○○○○○○○○

Applications
●○○

Summary
○

Cosmology

Cosmology

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- Plug into field equations of theory (either $f(\mathbb{G})$ or $f(\mathbb{G}, \mathbb{B})$)

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$$-f(\mathbb{G}) - 4f'(\mathbb{G})(\dot{H}(t) + 3H(t)^2) + 48H(t)^2 \dot{H}(t) f''(\mathbb{G}) = 2\kappa p(t)$$

with Hubble parameter $H(t) = \frac{\dot{a}(t)}{a(t)}$

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Introduction
○○

General Relativity
○○○○○

Gamma Squared Action
○○○○○

Modified Gravity
○○○○○○○○○

Applications
○○●

Summary
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- (Continuing work): Investigate these 'preferred coordinates'

Introduction
○○

General Relativity
○○○○○

Gamma Squared Action
○○○○○

Modified Gravity
○○○○○○○○○

Applications
○○○

Summary
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- Field equations no longer covariant (or divergence-free)
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- Coordinates play an important role